

AP CSP | Lesson 14: Conditional Loops

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: A

A follows the pre-check rule. B assumes a post-check loop; C invents an extra run; D reverses the stopping condition.

Q2. Answer: C

C: the states are 2, 4, 6, 8 and the loop stops at 8. A is before the final update; B assumes exact arrival; D adds once after stopping.

Q3. Answer: D

D: only odd values are reached. A and C stop without equality; B assumes the sequence reaches an even value.

Q4. Answer: B

B matches the sentinel contract. A includes the stopping marker as data; C rejects valid zero; D is unnecessary and may skip data.

Q5. Answer: A and C

A leaves count at 0. C moves below 0, away from 3. B stops immediately; D stops earlier. Both A and C are required.

Q6. Answer: C

C fixes five executions. A cannot progress when initially false; B runs zero times; D runs six times under the stated start/update.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: (a) 1, 3, 5. (b) 7. (c) 3. (d) 4.

1 mark: Gives outputs 1, 3, 5 in order.

1 mark: Gives final $n = 7$.

1 mark: Gives three body executions.

1 mark: Gives four checks, including the final true check.

Q8. 4 marks

Answer: $\text{charge} \leftarrow 20$

REPEAT UNTIL($\text{charge} \geq 65$)

```
{  
  charge  $\leftarrow$  charge + 15  
}
```

Final charge: 65. Body runs: 3.

1 mark: Initializes charge to 20 before the loop.

1 mark: Uses stop $\text{charge} \geq 65$ and update +15 inside the loop.

1 mark: Gives final charge 65.

1 mark: Gives three body executions.

Q9. 4 marks

Answer: (a) x stays 2, so the condition stays false. (b) $x \leftarrow x + 1$. (c) 2, 3, 4. (d) Start $x = 5$: 0 body executions.

1 mark: Connects unchanged $x = 2$ to a condition that stays false.

1 mark: Adds $x \leftarrow x + 1$ after DISPLAY in the body.

1 mark: Gives repaired outputs 2, 3, 4.

1 mark: Tests any starting $x \geq 5$ and predicts zero body executions.

Q10. 4 marks

Answer: Total 5; two body executions; three INPUT calls. Zero is valid data, while -1 is the stopping marker and is not added.

1 mark: Gives total 5.

1 mark: Gives two body executions.

1 mark: Gives three INPUT calls.

1 mark: Explains valid zero versus an excluded sentinel.

Q11. 4 marks

Answer: Equality never holds because n stays odd. $\geq$ ends at 7 after three updates. It fits “reach or exceed 6.”

1 mark: Explains that $n = 6$ is unreachable from the odd sequence.

1 mark: States that $n \geq 6$ stops at $n = 7$.

1 mark: Gives three body executions.

1 mark: States a reach-or-exceed requirement rather than exact equality.

Q12. 4 marks

Answer: $\text{count} \leftarrow 0$; stop at $\text{count} \geq 3$. Body: DISPLAY("Ready"), then $\text{count} \leftarrow \text{count} + 1$. Checks see 0, 1, 2 (false) then 3 (true).

1 mark: Initializes count consistently, such as 0.

1 mark: Uses a matching stopping test, such as $\text{count} \geq 3$.

1 mark: Includes DISPLAY and exactly one consistent count update in the body.

1 mark: Gives a trace proving three body executions.